NOT MEASUREMENT SENSITIVE
MIL-P-85856/35
7 December 1988

MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS
ALUMINA-TITANIA, TYPE II, CLASS 1, COMPOSITION E

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

| $Al_2O_3$ | $SiO_2$ | $TiO_2$ | $Fe_2O_3$ | Others (1) | |
|---|---|---|---|---|---|
| | | | | Each | Total |
| Bal | 3.0 max. | 2.0<br>4.1 | 2.0 max. | - | 1.0 max. |

(1) Determination not required for routine acceptance.

Particle Size/Distribution

| | |
|---|---|
| +325 Mesh (+45 µm) | 5.0 max. |
| -325 Mesh (-45 µm) | 95.0 min. |
| -15 µm | 10.0 max. |

Sprayed Coating Properties

| | |
|---|---|
| * Bond Strength | 13.8 MPa (2000 psi) min. |
| Hardness | Rockwell 15N, 85 min. |
| Microstructure | See paragraph 4.5.4.1.2<br>Porosity - 15% max.<br>Unmelted Particles - 5% max. |

* As applied over a NiAl bond coat.

AMSC N/A
FSC 3439
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Deposit Efficiency and Deposition Rate

70% min for $N_2/H_2$ @ 7.0 lb/hr
70% min for $A/H_2$ @ 7.0 lb/hr

Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

| | |
|---|---|
| General Electric Company | A50TF87, Class A |
| Rolls-Royce, Ltd. | MSRR 9507/36 |

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/35)



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